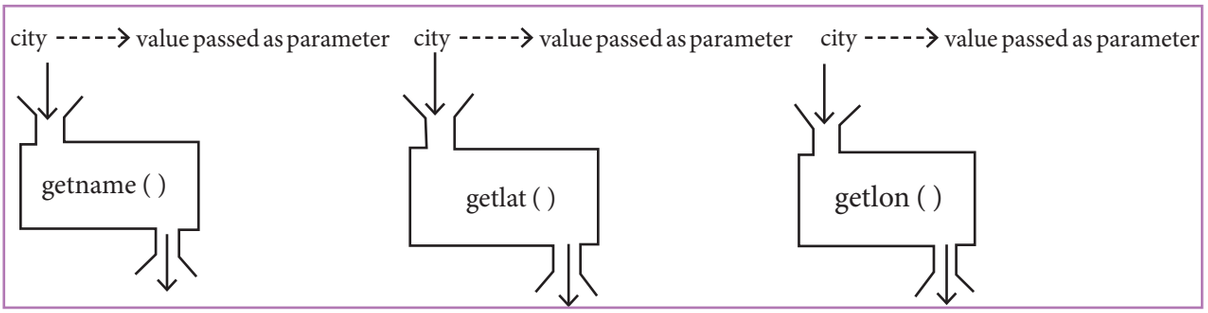


Now let us consider one more example to identify the constructor and selector for a slope.Read - - as comments.

```
- - constructor
makepoint(x, y):
    return x, y
- - selector
 xcoord(point):
    return point[0]
- -selector
ycoord(point):
    return point[1]
```

Note

Data abstraction is used to define an Abstract Data Type (ADT), which is a collection of constructors and selectors. Constructors create an object, bundling together different pieces of information, while selectors extract individual pieces of information from the object.

2.4 Representation of Abstract datatype using Rational numbers

The basic idea of data abstraction is to structure programs so that they operate on abstract data. That is, our programs should use data in such a way, as to make as few assumptions about the data as possible. At the same time, a concrete data representation is defined as an independent part of the program.

Note

A concrete data type is a data type whose representation is known.

Any program consist of two parts. **The two parts of a program are, the part that operates on abstract data and the part that defines a concrete representation**, is connected by a small set of functions that implement abstract data in terms of the concrete representation. To illustrate this technique, let us consider an example to design a set of functions for manipulating rational numbers.

Example

A rational number is a ratio of integers, and rational numbers constitute an important sub-class of real numbers. A rational number such as 8/3 or 19/23 is typically written as:

<numerator>/<denominator>

where both the <numerator> and <denominator> are placeholders for integer values. Both parts are needed to exactly characterize the value of the rational number. Actually dividing integers produces a float approximation, losing the exact precision of integers.

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